

Machine Learning Applications

Winter Semester 2022/2023 – Lecture 7

Jan Hämmelmann, M.Sc. & Prof. Dr.-Ing. Joachim Metternich | PTW



Data Challenge Kick-Off

Topics of Today

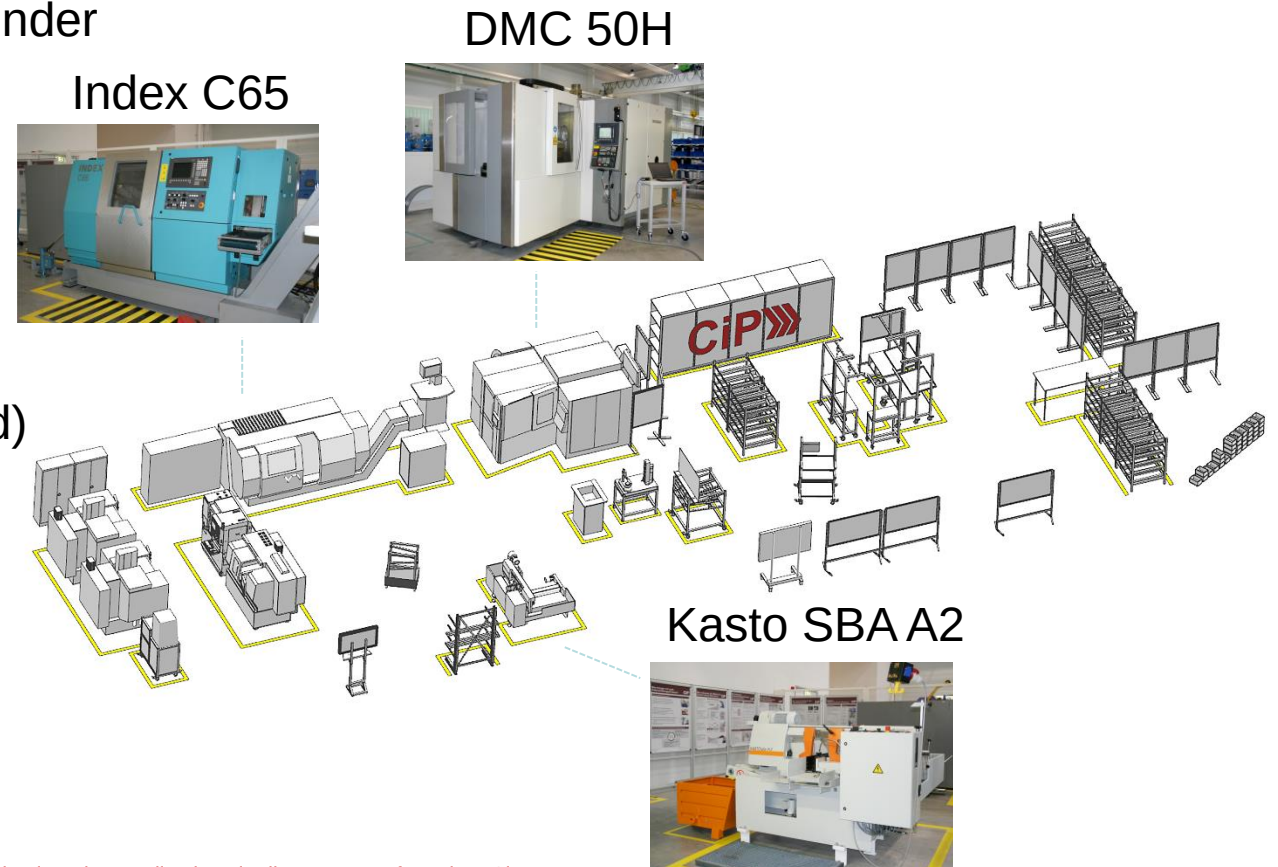
1. CiP-DMD: Dataset
2. Data Challenge Task

Data Challenge Kick-Off

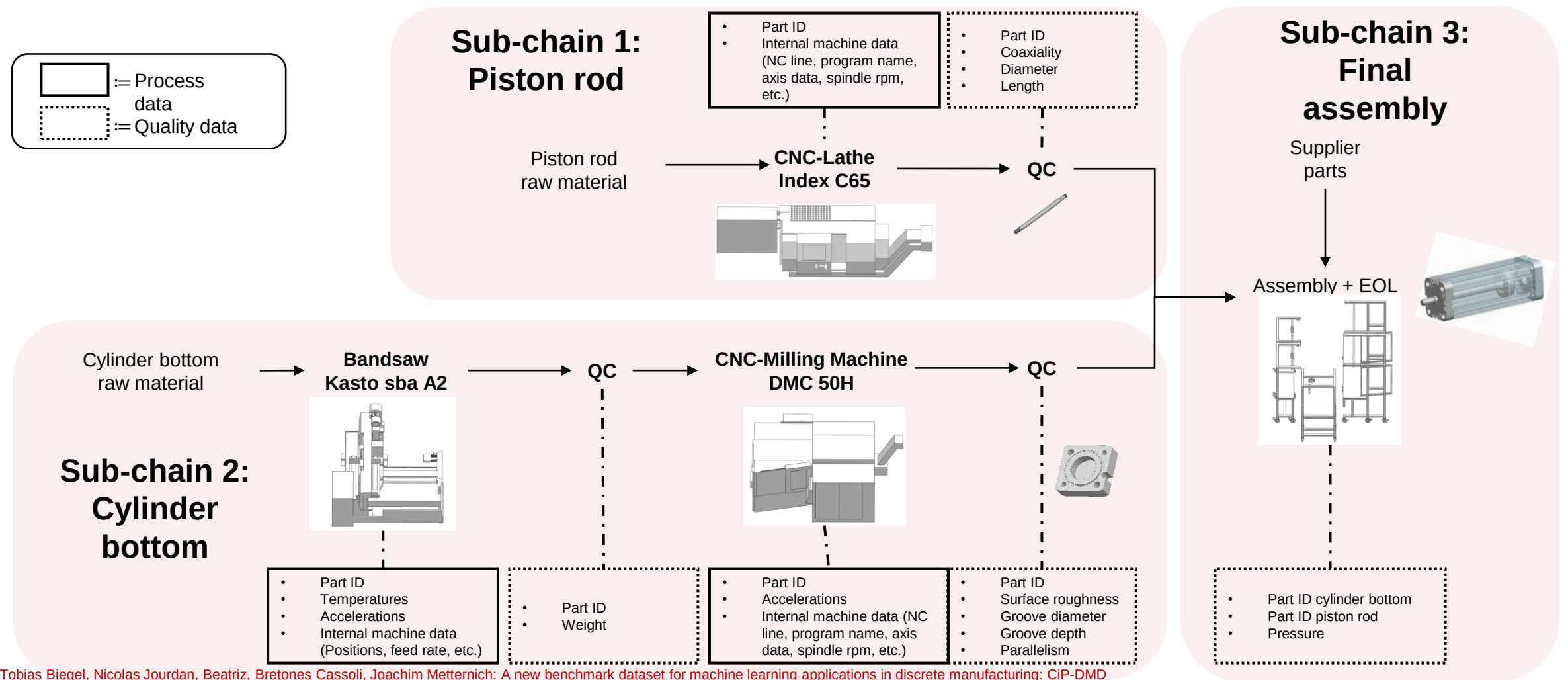
CIP-DMD: DATASET

CiP-DMD: Dataset Introduction

- CiP-DMD: **C**enter for industrial **P**roductivity – **D**iscrete **M**anufacturing **D**ataset
- Recorded in the process learning factory CiP at TU Darmstadt (November 2022 – June 2023)
- Multistep production process of a pneumatic cylinder
 - Sawing
 - Milling
 - Turning
 - Final Assembly
- Labelled data for 847 pneumatic cylinders
 - 697 normal
 - 150 anomalous (139 planned, 11 unplanned)



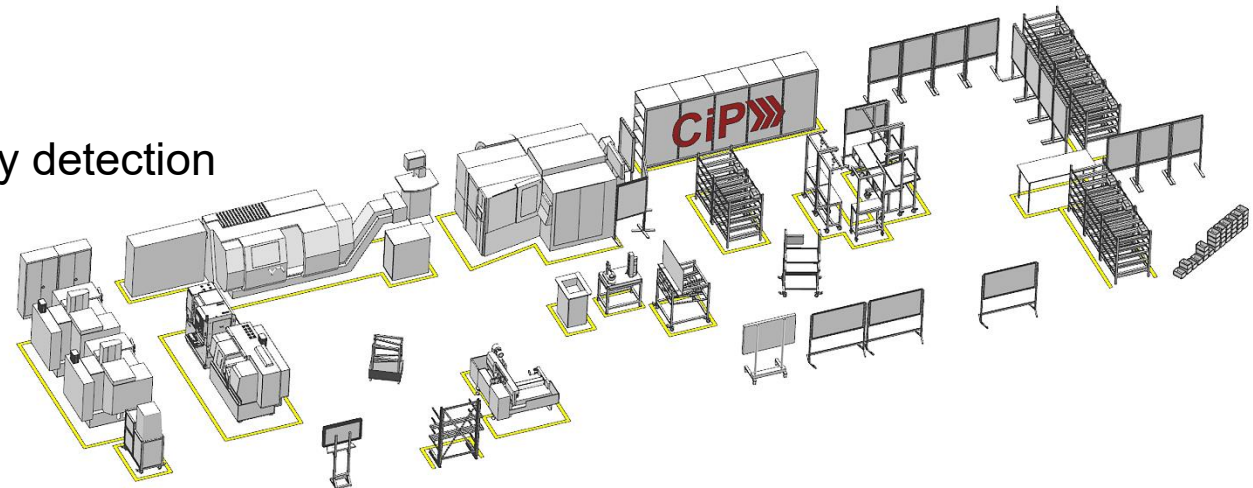
CiP-DMD: Manufacturing chain



Tobias Biegel, Nicolas Jourdan, Beatriz, Bretones Cassoli, Joachim Metternich: A new benchmark dataset for machine learning applications in discrete manufacturing: CiP-DMD

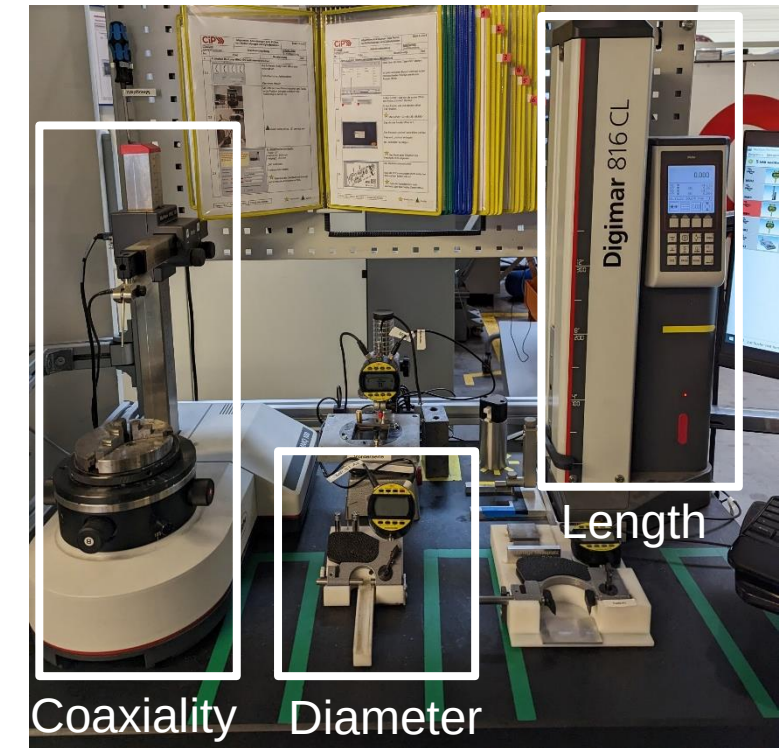
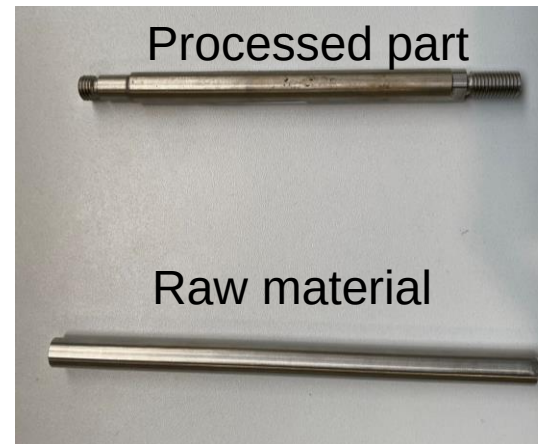
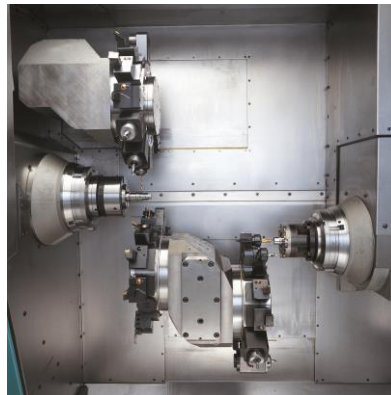
Motivation

- Badly manufactured sub-parts can lead to problems later in the manufacturing process
 - Mounting cannot take place if holes are drilled incorrectly
 - Seals do not close airtight if tolerances are not maintained
 - Cylinder rod cannot be moved if diameter is incorrect
- Anomalies in the production should be detected as soon as possible in the production process
- Leads to:
 - Faster correction of problems
 - Less scrap
 - Higher Quality
- Goal of dataset:
 - Product tracability
 - Record machining data → Can be used for anomaly detection



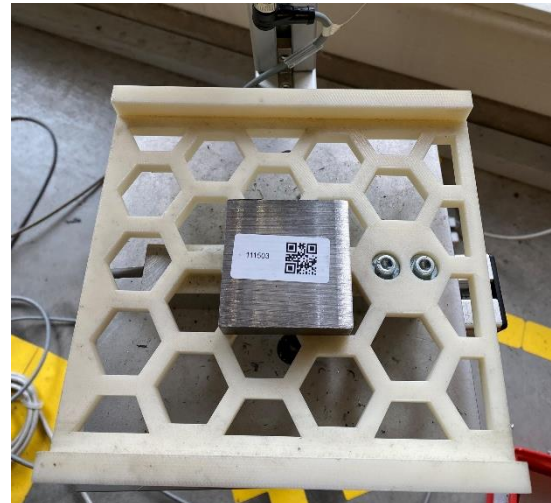
Sub-chain 1: Piston rod

- **Material:** Stainless steel
- **Process:** Turning (847 normal, 0 anomalous)
- **Processing time of a machine cycle:** 436s (four rods per cycle)
- **Process data:**
 - Source: Siemens Brownfield Connectivity (BFC) Gateway
 - Protocol: MQTT (5Hz sampling rate)
 - Internal machine control data, e.g., NC line, spindle current, axis data
- **Quality data:**
 - Hexagon/Q-DAS quality control software + Mahr measurement devices
 - Coaxiality, Diameter, Length



Sub-chain 2: Cylinder bottom

- **Material:** Steel
- **Process:** Sawing (748 normal, 99 anomalous - too shortly cut)
- **Processing time of a machine cycle:** 143s (per part)
- **Process data:**
 - Source: External sensors
 - Protocol: OPC UA (5Hz sampling rate)
 - Temperatures, accelerations, etc.
- **Quality data:**
 - Source: OPC UA
 - Scale
 - Weight of the sawn part



Sub-chain 2: Cylinder bottom

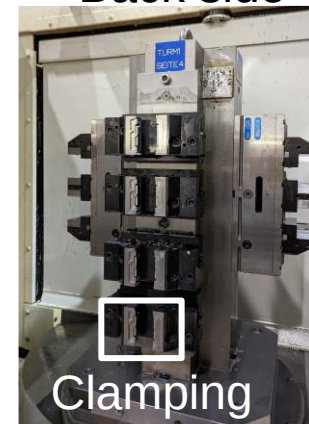
- **Process:** Milling (697 normal, 150 anomalous – 139 planned, 11 unplanned)
- **Processing time of a machine cycle:** 312s (front side: 199s, back side: 113s)
- **Process data:**
 - Source: External sensors (triaxial accelerometer)
 - National Instruments cards (2.5kHz sampling rate)
 - Accelerations in x, y, z
 - Source: Siemens Brownfield Connectivity (BFC) Gateway
 - Protocol: MQTT (5Hz sampling rate)
 - Internal machine control data, e.g., NC line, spindle current, axis data



Front side

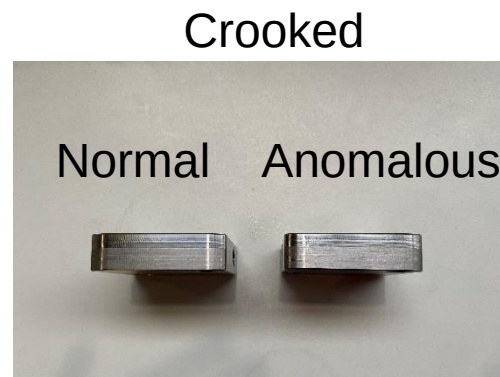
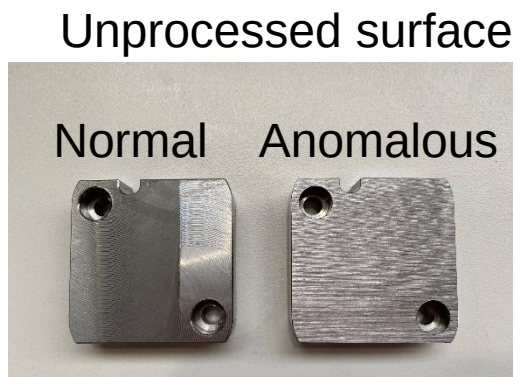
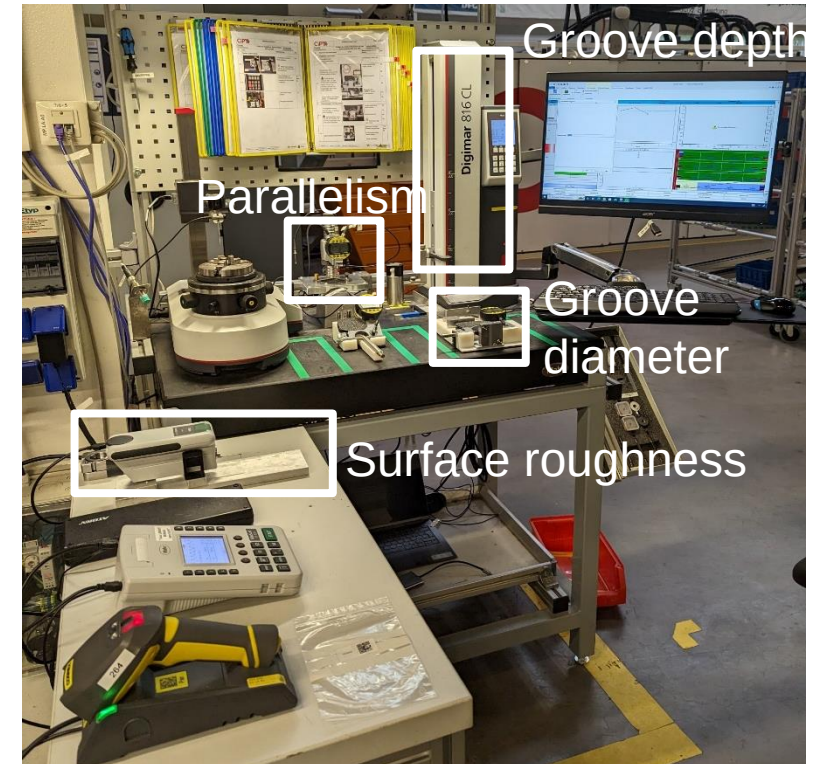


Back side



Sub-chain 2: Cylinder bottom

- **Quality data:**
 - Hexagon/Q-DAS quality control software + Mahr measurement devices
 - Surface roughness, parallelism, groove depth, groove diameter
- **Planned anomalies (139):**
 - Raw – Semi: Unprocessed surface (99) → Part too short – affects surface roughness
 - Semi – Finished: Crooked (40) → Part falsely clamped – affects parallelism
- **Unplanned anomalies (11):** Broken tool, wrong machine settings



Sub-chain 3: Final assembly

- **Process:** Assembly (697 normal, 150 anomalous)
- **Quality data:**
 - Source: OPC UA
 - Load cell
 - Final pressure test

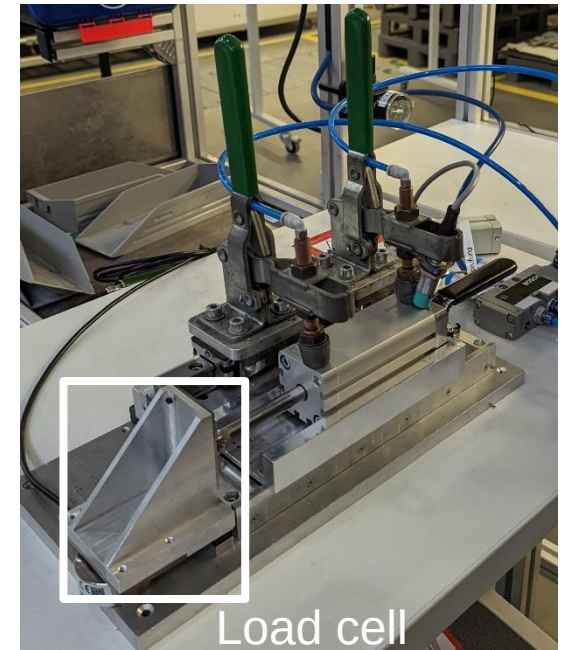
Assembly station 1



Assembly station 2



Pneumatic cylinder components



Tobias Biegel, Nicolas Jourdan, Beatriz, Bretones Cassoli, Joachim Metternich: A new benchmark dataset for machine learning applications in discrete manufacturing: CiP-DMD

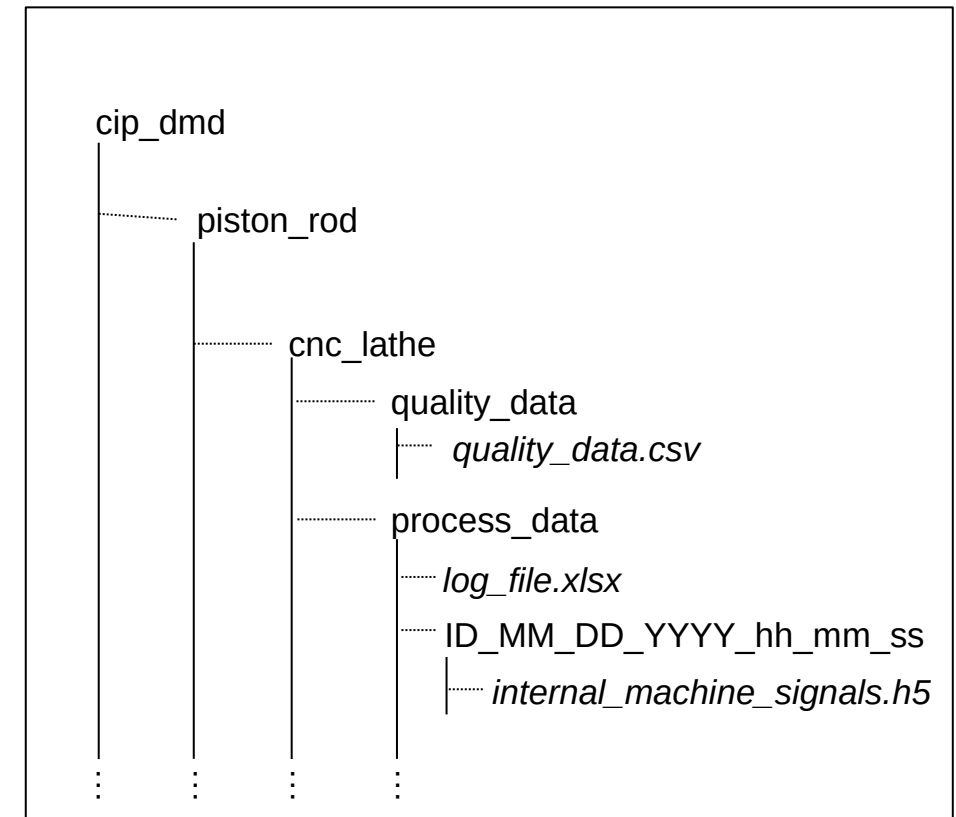
Traceability concept and dataset structure

- QR codes (ID) + Time Stamps represent anchor points to map the data to the physical parts

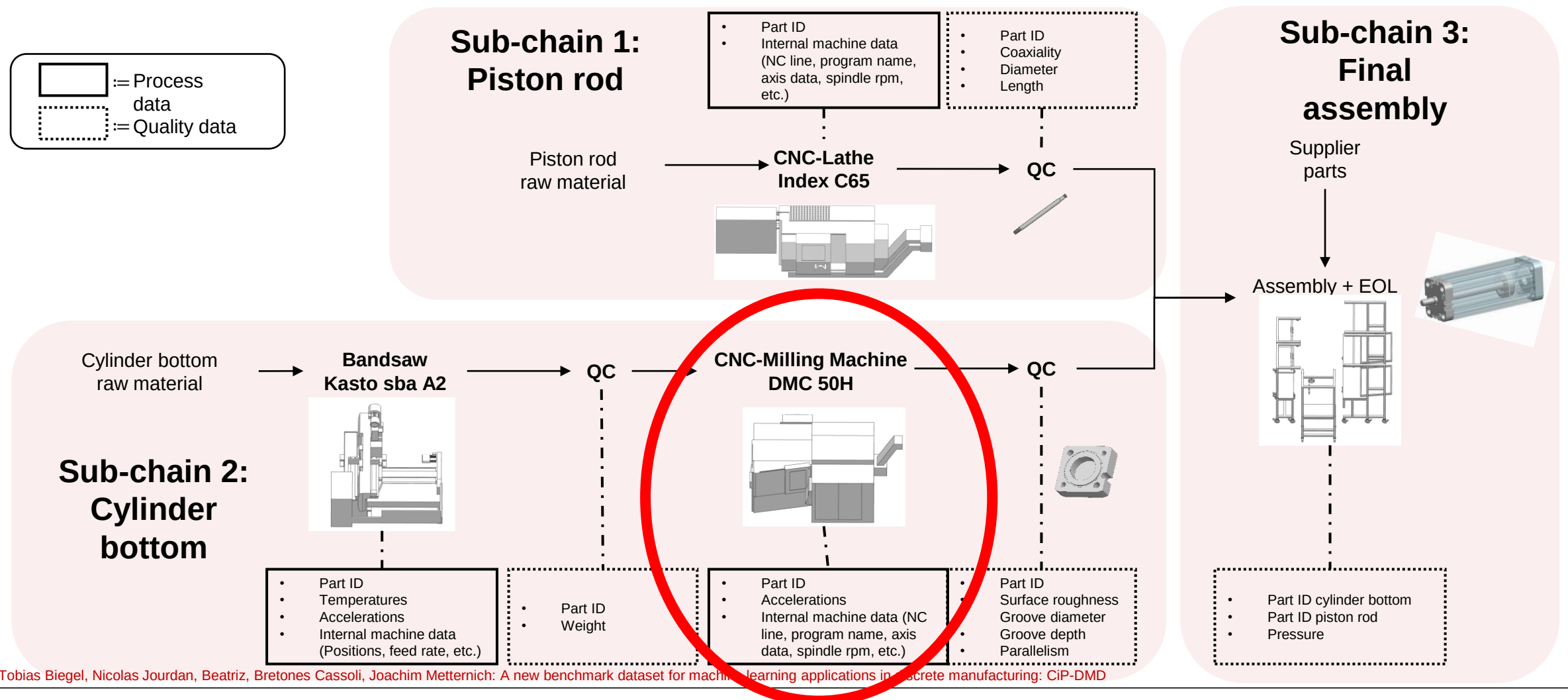
CiP-DMD structure

1 = Cylinder
bottom
2 = Piston rod

1 001 01 → ID
Batch



CiP-DMD: Manufacturing Chain



Tobias Biegel, Nicolas Jourdan, Beatriz, Bretones Cassoli, Joachim Metternich: A new benchmark dataset for machine learning applications in concrete manufacturing: CiP-DMD

Meta Data for cylinder bottom: **cylinder_bottom/meta_data.json**

```
{
  "part_type": "cylinder_bottom",
  "part_id": "100101",
  "component_ids": [],
  "process_data": [
    {
      "data_paths": [
        "cylinder_bottom/saw/process_data/100101_8_16_2022_8_59_47/internal_machine_signals.h5"
      ],
      "start_time": 1660633187.003,
      "end_time": 1660633295.504,
      "anomaly": "0",
      "name": "saw"
    },
    {
      "data_paths": [
        "cylinder_bottom/cnc_milling_machine/process_data/100101_11_29_2022_12_30_29/frontside_internal_machine_signals.h5",
        "cylinder_bottom/cnc_milling_machine/process_data/100101_11_29_2022_12_30_29/frontside_timestamp_process_pairs.csv",
        "cylinder_bottom/cnc_milling_machine/process_data/100101_11_29_2022_12_30_29/frontside_external_sensor_signals.h5",
        "cylinder_bottom/cnc_milling_machine/process_data/100101_11_29_2022_12_30_29/backside_timestamp_process_pairs.csv",
        "cylinder_bottom/cnc_milling_machine/process_data/100101_11_29_2022_12_30_29/backside_external_sensor_signals.h5",
        "cylinder_bottom/cnc_milling_machine/process_data/100101_11_29_2022_12_30_29/backside_internal_machine_signals.h5"
      ],
      "start_time": 1669721429.979765,
      "end_time": 1669722386.925569,
      "anomaly": 0,
      "name": "cnc_milling_machine"
    }
  ],
}
```

- **Anomaly:**
 - 0 : Normal process
 - 1 : Raw cutting material was badly aligned at the saw
 - 2 : Part was unevenly clamped in the milling jig
 - 3 : Miscellaneous errors happened during the process that are not visible in process data
- **Sensor Data**
 - internal_machine_signals: Information about processing steps
 - external_sensor_signals: Triaxial accelerometer to measure vibrations at spindle



Data Challenge Kick-Off

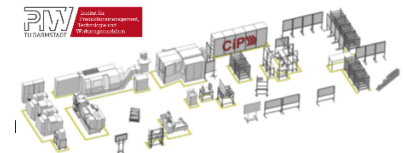
DATA CHALLENGE TASK

Task

Main objective

Detect anomalies of the cylinder-bottom parts using processing data from the CNC milling machine. The output of your analysis shall be a binary classification of the cylinder bottom parts:

- *False*: no anomaly (anomaly label 0)
- *True*: detected anomaly (anomaly labels 1, 2 and 3)



The Machine Learning Applications Data Challenge 23/24

Topic: Anomaly detection in pneumatic cylinder production

Duration: 11.12.2023 – 12.02.2024

Background

Within the framework of Industry 4.0, manufacturing machines are increasingly being equipped with sensors that automatically generate and store data in the production process. The analysis of the data can be a driver to better understand the production process and enable predictions about product quality. One key aspect is the detection and assessment of anomalies which can lead to problems in later stages of the production process.

Task Definition

Within this task, you are going to examine data from the production of pneumatic cylinders. The cylinders are assembled from several parts with one of them being the cylinder bottom. A two-stage production process is used to manufacture the cylinder bottoms. First, a piece of raw material is cut from a square bar of steel using a bandsaw. Then, the sawn parts are finished in a CNC milling machine.

You dispose of process data from the CNC milling machine. This includes data from external accelerometer sensor data, as well as internal machine control data. Given this information, your task is to analyze the given data base and detect anomalies that were introduced into some of the cylinder bottom parts.

Main objective

Detect anomalies of the cylinder-bottom parts using processing data from the CNC milling machine. The output of your analysis shall be a binary classification of the cylinder bottom parts:

- *False*: no anomaly (anomaly label 0)
- *True*: detected anomaly (anomaly labels 1, 2 and 3)

The F1-score of the *True* class will be used for evaluation of results.

In order to achieve this objective, follow the listed sub-tasks:

- Explore and analyze the process data from the milling machine
- Extract meaningful features for your anomaly detection model
- Develop a machine learning model for anomaly detection
- Evaluate the errors, uncertainty and robustness of the model

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THANK YOU VERY MUCH FOR YOUR ATTENTION



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